Musanze Harvest Intelligence Service

SWE 3513 — Individual Assignment 2

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Verification code: AI-A2-7201 | Random seed: 7201

This report is generated from the current run. Verify all outputs and add your own interpretation before submission.

1. Identity and decision problem

The stakeholder is a Musanze potato cooperative field officer. The prototype estimates harvest quantity, predicts a binary high-risk label, and groups farms with similar input conditions. The decision-support output is not a substitute for agronomist judgement.

2. Data and reproducibility

The data are synthetic. CSV SHA-256: d013ff87a7c7f2778ea9069b4ca170fdf37aa928c6840eff31f5e040ff2067db. Rows before duplicate removal: 269; duplicates removed: 3; rows after cleaning: 266. Missing values found after deduplication: {"soil_moisture_pct": 13, "fertilizer_kg": 13}.

3. Pipeline design and leakage controls

A fixed 75/25 train/test split uses the personal seed. Numeric imputation/scaling and categorical imputation/one-hot encoding are contained in scikit-learn pipelines and fitted on training data for regression and classification. Yield and risk targets are excluded from model input. Clustering uses input features only; its selected final model is refit on all transformed input rows after K selection on the training split.

4. Regression

Model	MAE	RMSE	R^2
mean_baseline	2.312	2.814	-0.001
ridge	0.879	1.111	0.844
random_forest	1.075	1.376	0.761

Selected regression model: ridge. Selection rule: Lowest held-out RMSE among the two candidate models.

Interpretation prompt: inspect artifacts/test_predictions.csv, find a row with a large absolute yield_error, and explain that error in your own words. Do not claim causes that the synthetic features cannot establish.

5. Classification

Model	Precision	Recall	F1	ROC-AUC
majority_baseline	0.463	1.000	0.633	0.500
logistic_regression	0.857	0.968	0.909	0.986
random_forest	0.906	0.935	0.921	0.959

Selected classifier: random_forest. Highest held-out F1 among the two candidate classifiers.

Interpretation prompt: use the confusion matrix and test_predictions.csv to discuss one false positive or false negative. Explain the operational consequence cautiously.

6. Clustering

Silhouette scores: {"2": 0.15722136686422433, "3": 0.14216836367531155, "4": 0.14550050070982423, "5": 0.14237166349964028, "6": 0.1417222205951469}. Selected K=2. Highest silhouette score among K=2 through K=6 on the training split.

Cluster 0: n=134, mean rainfall=1011.88, mean soil moisture=51.14, mean fertilizer=135.62, mean field size=2.17, support rate=0.0. Write a grounded operational interpretation after inspecting this profile.

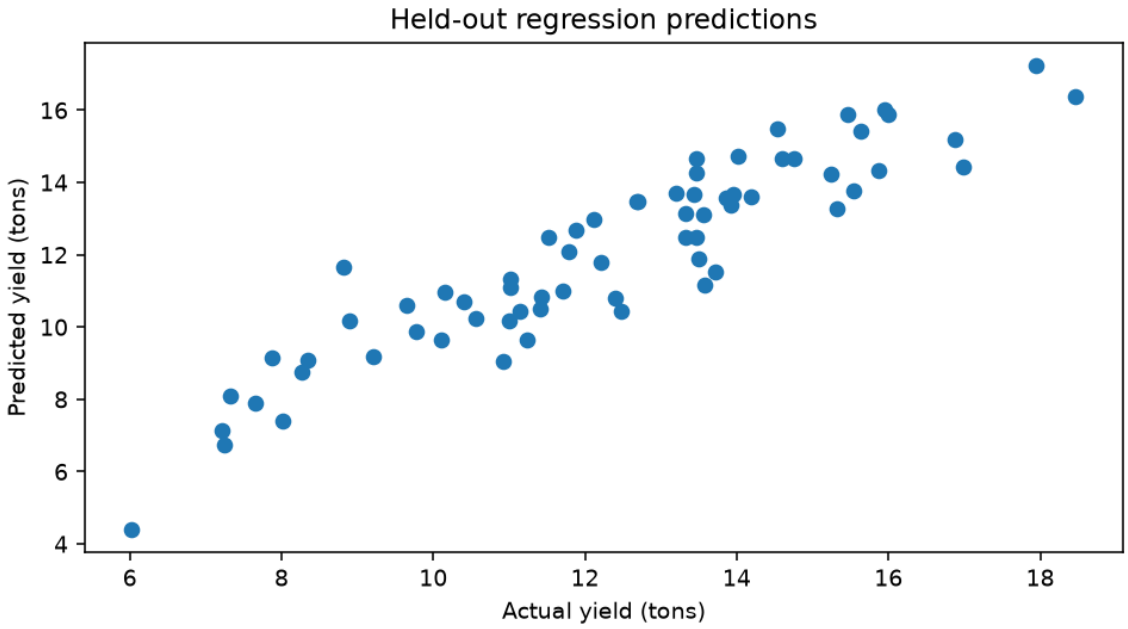
Cluster 1: n=132, mean rainfall=988.64, mean soil moisture=50.88, mean fertilizer=126.63, mean field size=2.19, support rate=1.0. Write a grounded operational interpretation after inspecting this profile.

7. API, testing, and responsible AI

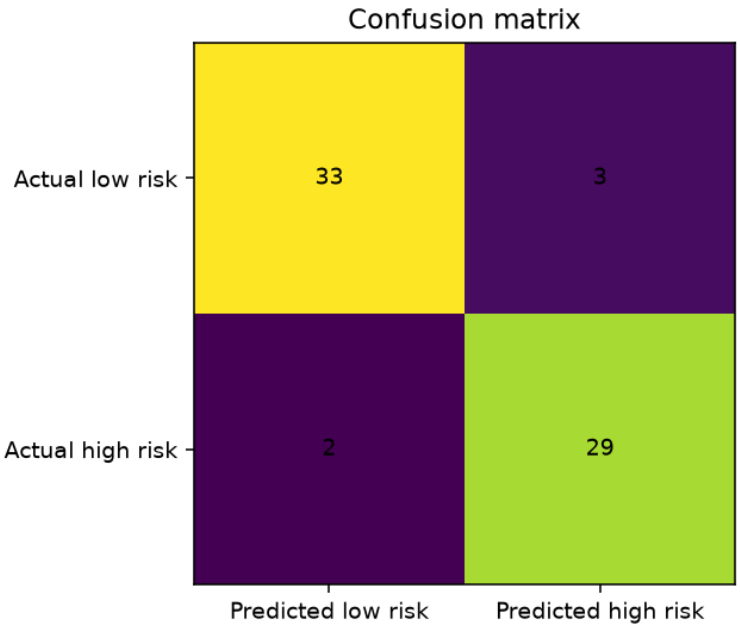
The service exposes GET /health, GET /model-info, and POST /predict. Input validation rejects missing fields, invalid categories, and values outside expected ranges. Run pytest -q and insert the actual passing-test count and evidence in your final version.

Limitations: synthetic data do not represent measured Musanze farms; the generated risk label is defined from synthetic yield per hectare; model performance is not evidence of real-world performance. Do not use this tool to decide credit, insurance, or public-service eligibility. Human review by a qualified agricultural expert is required. With real data, obtain consent and governance approvals, validate representativeness, evaluate fairness and calibration, and monitor drift.

Appendix: residuals.png



Appendix: confusion_matrix.png



Appendix: clusters.png

